Scalability & Future Plan Roadmap

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Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	SQLite single file database is designed for local testing.	Inhibits massive concurrent write throughput in production.	Medium
2	Manual data entry required to populate borrower loan registers.	Increases friction for users tracking multiple card balances.	High
3	Gemini API limits (rate and context constraints).	Could throttle prompt responses under heavy concurrent loads.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-threaded development local server bindings.	Transition backend container to Gunicorn with Uvicorn workers behind Nginx load balancers.
2	Data Storage	Local SQLite database file.	Migrate connection mappings to cloud PostgreSQL cluster supporting automated connection pooling.
3	Performance	Direct API requests blocking thread cycles.	Implement Redis caching layer for dashboard calculation caches and session storage.
4	Security	PBKDF2 SHA-256 secure hashes and JWT authorization.	Integrate Multi-Factor Authentications (MFA) and SSL/TLS endpoint encryptions.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Direct bank transaction synchronization (Plaid API integration).	Month 2	Eliminates manual entry; matches balances instantly.
Phase 3	Integrate credit score reporting bureau APIs.	Month 4	Shows real-time credit score restorations on the user dashboard.
Phase 4	Digital settlement portals linked to banking APIs.	Month 6	Enables borrowers to execute settlements digitally from the app.